Software Requirements Document

Enterprise IT Support Agentic RAG Copilot

Document Version	1.0.0 (IEEE 830 Conformance)
Standard	IEEE Std 830-1998 Software Requirements Specification
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Test Suite Status	38 of 38 Tests Passing (0.77s execution time)
Python Runtime	Python 3.11 / 3.12 / 3.13 Compatible
Packaging	Dockerized Multi-stage Container & Virtualenv

1. System Environment & Hardware Requirements

The software operates as a standalone microservice or containerized daemon capable of running on enterprise on-premises Linux/Windows hosts or AWS/GCP cloud virtual machines.

- **Host Operating System:** Windows 10/11, Ubuntu 22.04 LTS+, macOS Sonoma+, or Debian-based Docker container.
- **CPU & Memory:** Minimum: 2 vCPU cores, 4GB RAM. Recommended for production: 4 vCPU cores, 8GB RAM.
- **Disk Storage:** 500MB free disk space for application files, SQLite audit database, and uploaded documents.
- **Network Connectivity:** Outbound HTTPS (port 443) access to Groq API (api.groq.com), Pinecone API, and Tavily API.

2. Software Dependency Matrix

Package	Version	Technical Purpose in Copilot
fastapi	0.116.1	High-performance asynchronous backend API framework
uvicorn[standard]	0.35.0	Lightning-fast ASGI production web server
langgraph	1.1.10	StateGraph cyclic agentic decision orchestrator
langchain-groq	1.1.3	Integration client for Groq high-speed LLM inference
sentence-transformers	6.0.0	HuggingFace all-MiniLM-L6-v2 dense vector embedding model
pinecone	7.3.0	Pinecone Serverless vector database SDK
langchain-tavily	0.2.18	Real-time vendor and web search tool client
pytest	8.3.4	Automated unit, integration, and dataset validation test runner
httpx	0.28.1	FastAPI TestClient HTTP testing client

3. Verification, Validation & Test Matrix

The application includes an automated test suite guaranteeing regression-free updates and verified accuracy:

- **tests/test_dataset_validation.py (10 Tests):** Validates existence, UTF-8 integrity, and section policies of company_it_handbook.md and service_desk_runbook.md. Validates chunking boundaries and eval_dataset.json schema.
- **tests/test_api_endpoints.py (9 Tests):** Validates GET /api/health, UI rendering, POST /api/chat error handling, input validation (422), audit database persistence, and POST /api/ingest security authorization.
- **tests/test_evaluation_service.py (9 Tests):** Validates keyword coverage algorithms, trace route extraction, sample pass/fail thresholding, and mock evaluation reporting.
- **tests/test_workflow_units.py (10 Tests):** Unit tests all LangGraph nodes (routing, grading, rewriting) with external LLMs mocked for fast deterministic execution.

VERIFIED TESTING STATUS: Continuous Quality Verification: All 38 automated test cases execute cleanly in under 1 second (0.77s - 1.03s), making the test suite ideal for pre-commit hooks and CI/CD deployment pipelines.